



Mudabbr: An Intelligent Expense Management System Using OCR Technology



Mudabbr Application System



Under the supervision of

Dr. Najlaa Jannah

Prepared by

**Baylsan Hassan Almehdawi
Jood Mohammed Al-Hatemi
Rose Ibraheem Al-Hatemi
Fatima Mohammed Al-Barakati
Etimad Awwad Almalki**



Mudabbr Application System



Mudabbr



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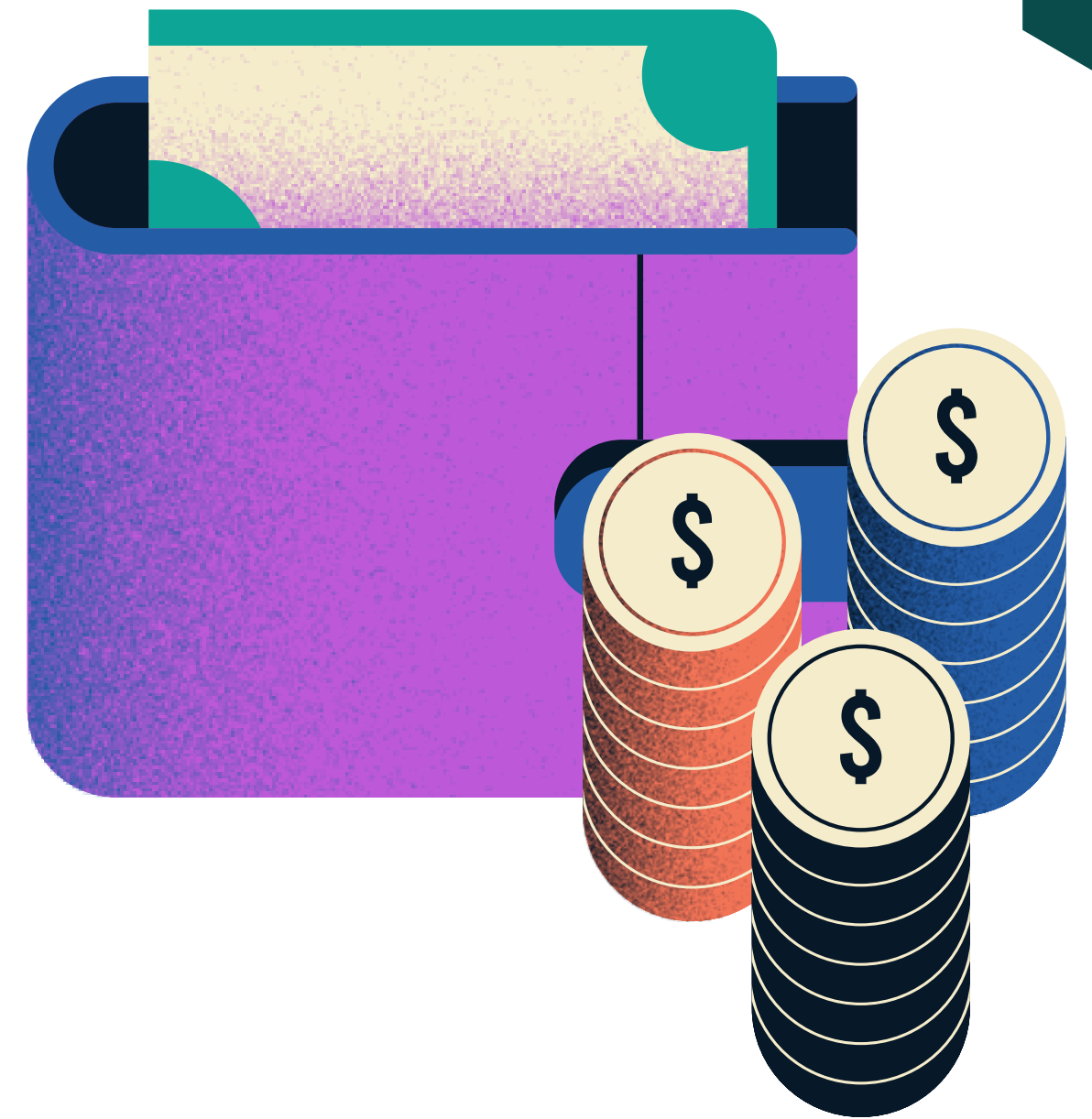
Introduction

Monthly commitments and expenses have increased, making it difficult for many people to track receipts and remember bill payment dates.

The Mudabbr app was developed as a smart solution to manage daily and monthly expenses efficiently.

The app uses OCR technology to scan bills and automatically extract data such as amount and date. Expenses are categorized into essential areas, and the system provides monthly reports and expense predictions.

With password and biometric authentication, Mudabbr offers a secure, smooth, and reliable financial management experience.





Project Purpose



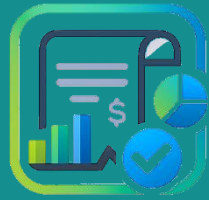
The automated receipt management and expense tracking tools in the Mudabbr application are set to revolutionize the way the user manages their daily and monthly spending.



Objectives

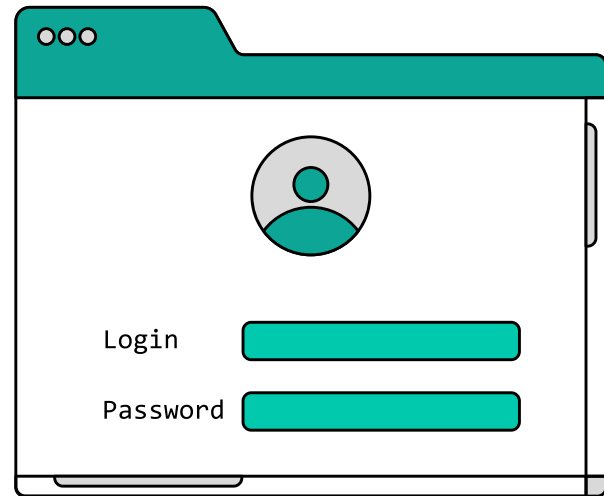
- 💰 Smart receipt scanning & auto data capture.
- 💰 Clear spending insights & budgeting categories.
- 💰 Monthly spending prediction & habit improvement.
- 💰 Yearly highlights of top purchases & behavior awareness.
- 💰 Promote smart, and budgeting habits.
- 💰 Timely alerts & bill payment reminders.
- 💰 Easy-to-use financial app.
- 💰 Automate tasks to save time & improve planning.





Mudabbr

System Implementation



System Database

mudabbr-9e3f423f

Realtime Database

Need help with Realtime Database? Ask Gemini

Data

Rules

Backups

Usage

Extensions

Protect your Realtime Database resources from abuse, such as billing fraud or phishing

Configure App Check

https://mudabbr-9e3f423f-default-rtdb.firebaseio.com

https://mudabbr-9e3f423f-default-rtdb.firebaseio.com/

Expense

administrator

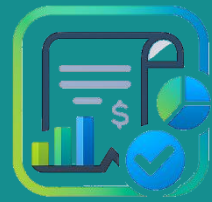
bills

boolean found = false;

income

notification

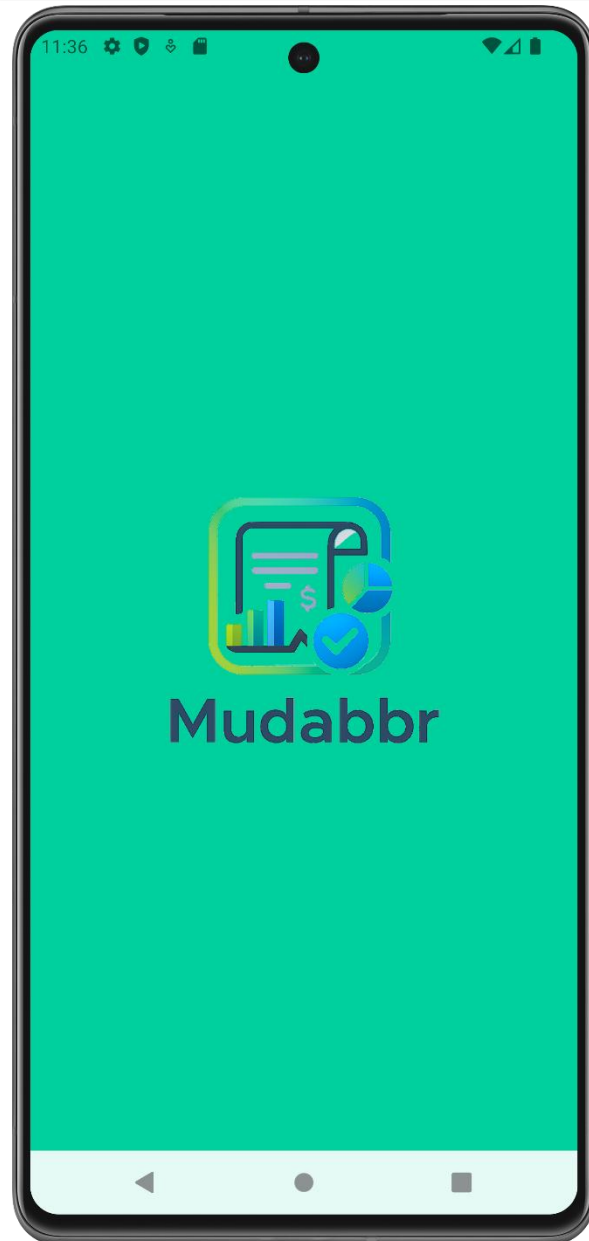
users



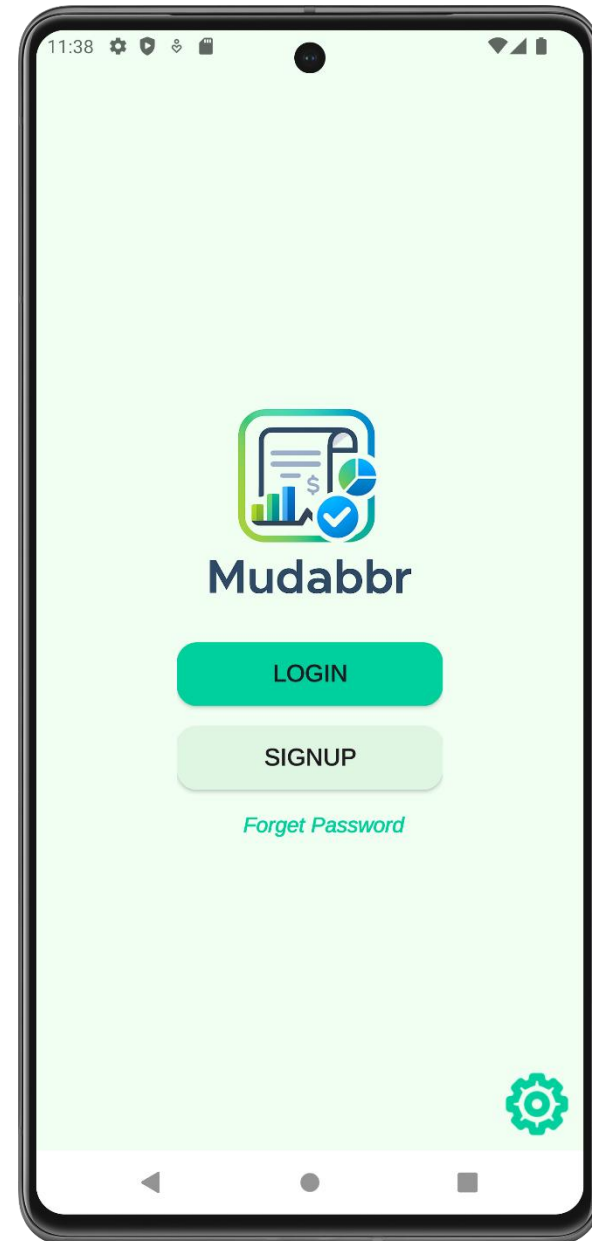
Mudabbr

System Implementation

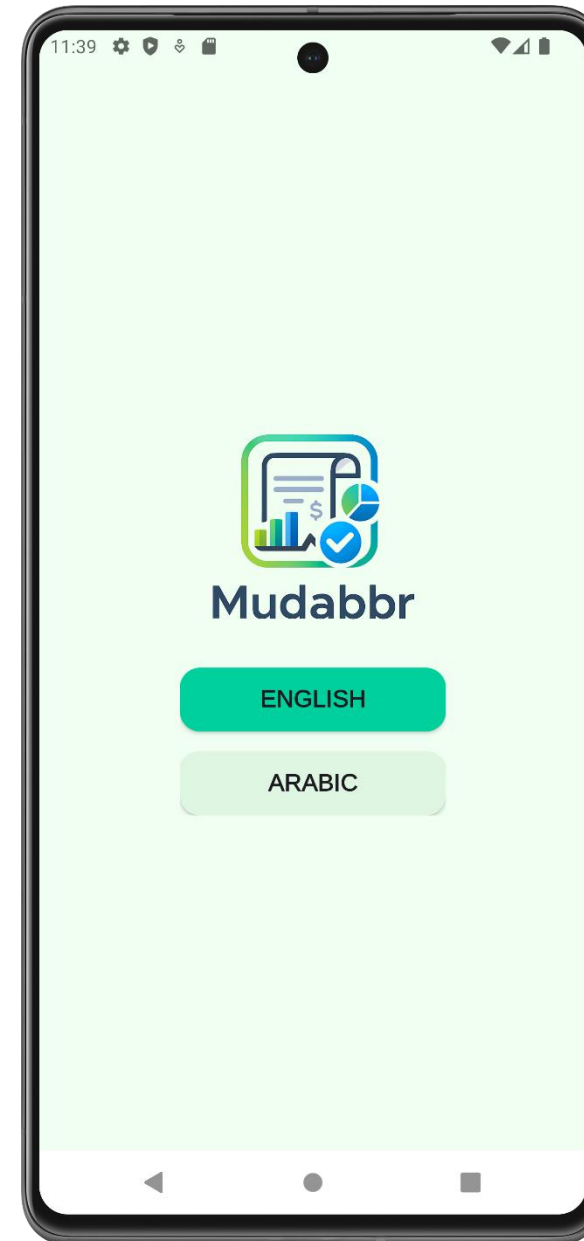
Splash interface



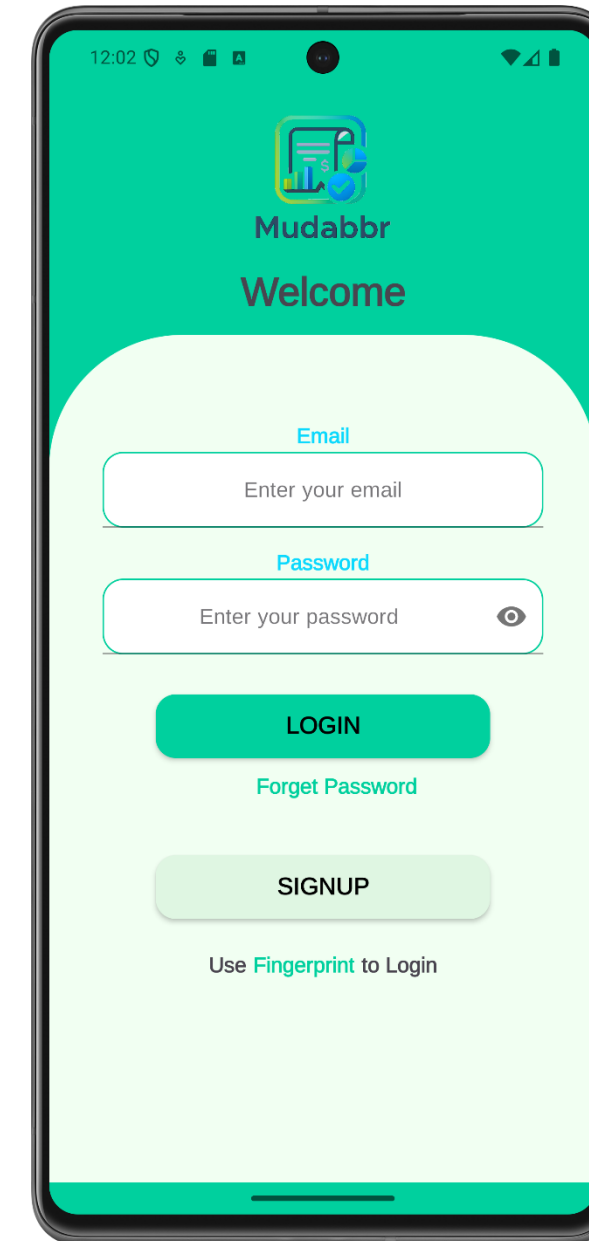
Start interface



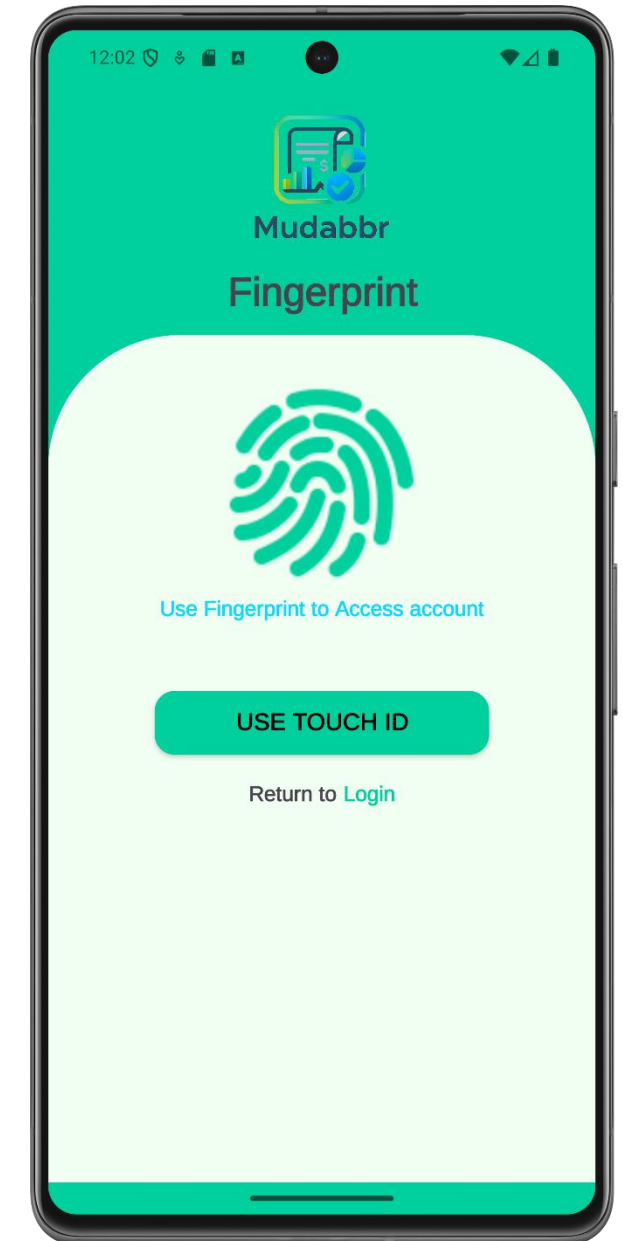
Settings interface

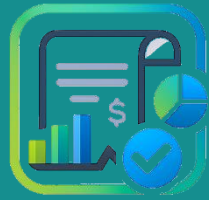


Login interface



Fingerprint





Mudabbr

System Implementation

Signup interface



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Create Account

Full Name
Enter your name

Email
Enter your email

Mobile Number
Enter your Phone

Date of Birth
dd-mm-yyyy

Password
Enter your password

By continuing, you agree to Terms of Use and Privacy Policy.

SIGNUP

Already have an account? [Login](#)

Forget Password 1



Mudabbr

Forget Password

Reset Password
To reset password, type your email and we will send secret code to your email.

Email
Enter your email

SUBMIT

NEXT

Forget Password 2



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Security Pin

Enter Security Pin

X X X X X X

SUBMIT

SEND AGAIN

Forget Password 3



Mudabbr

New Password

New Password
Enter your password

Confirm New Password
Confirm New Password

RESET PASSWORD

Edit Profile Interface



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Edit Profile

Full Name
Enter your name
System Manager

Mobile Number
Enter your Phone
0562323228

Password
Enter your password

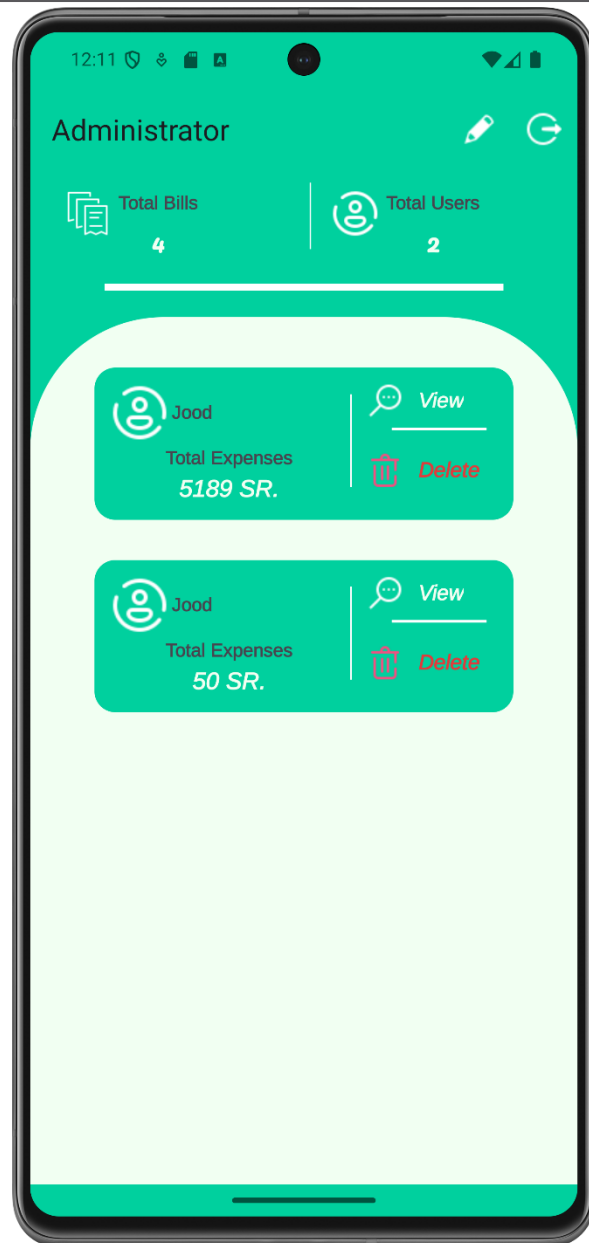
SAVE



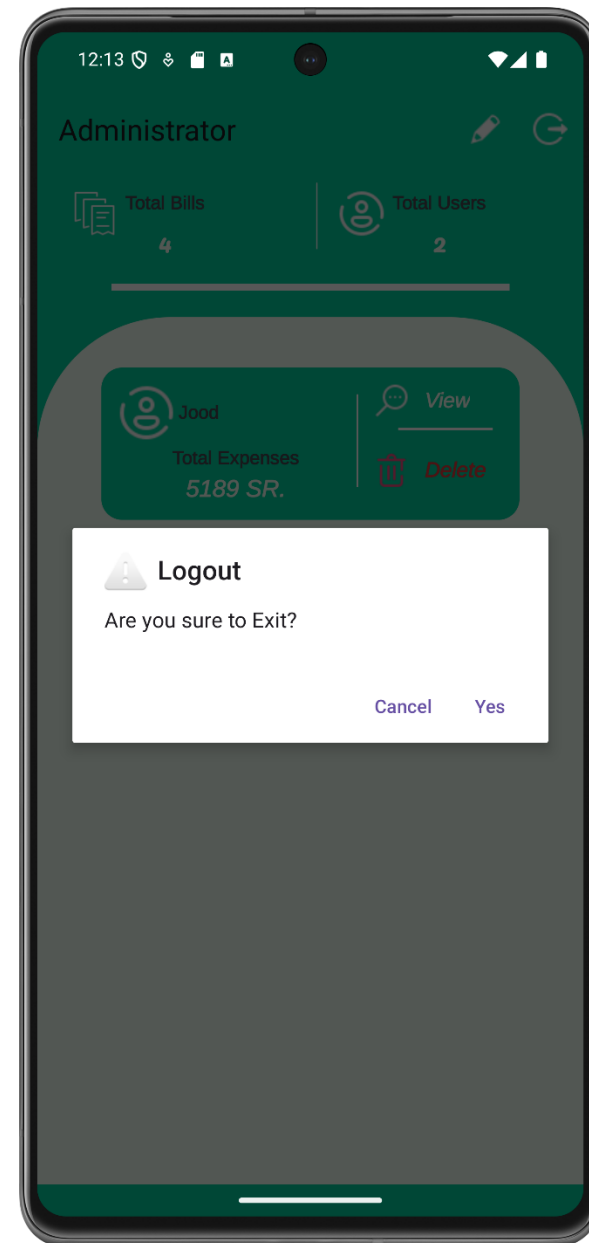
Mudabbr

System Implementation

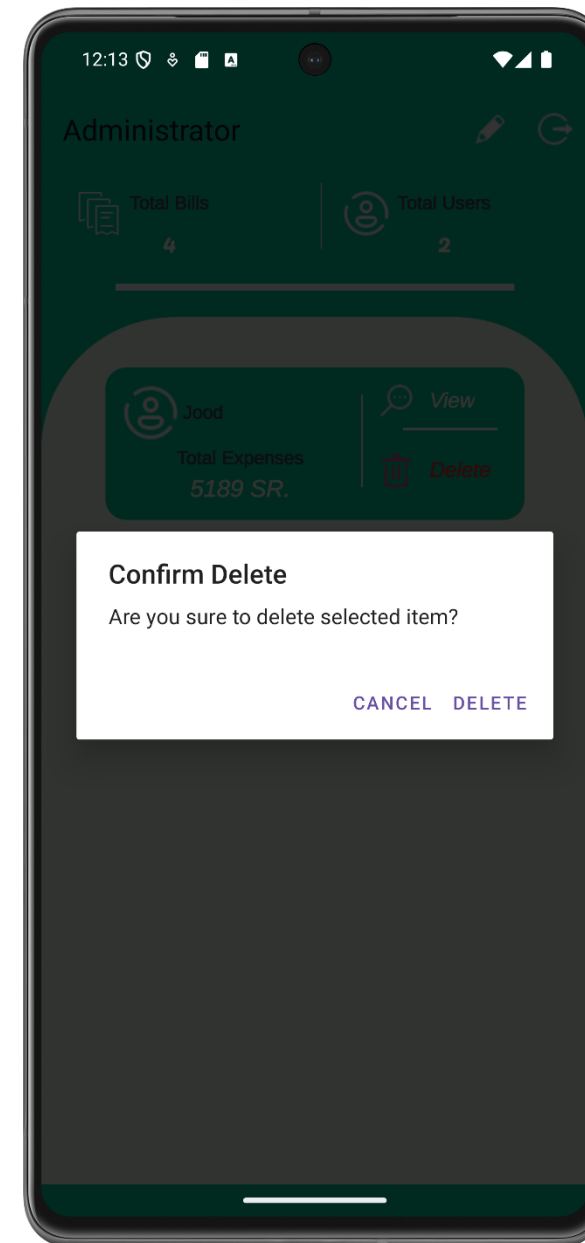
Admin interface



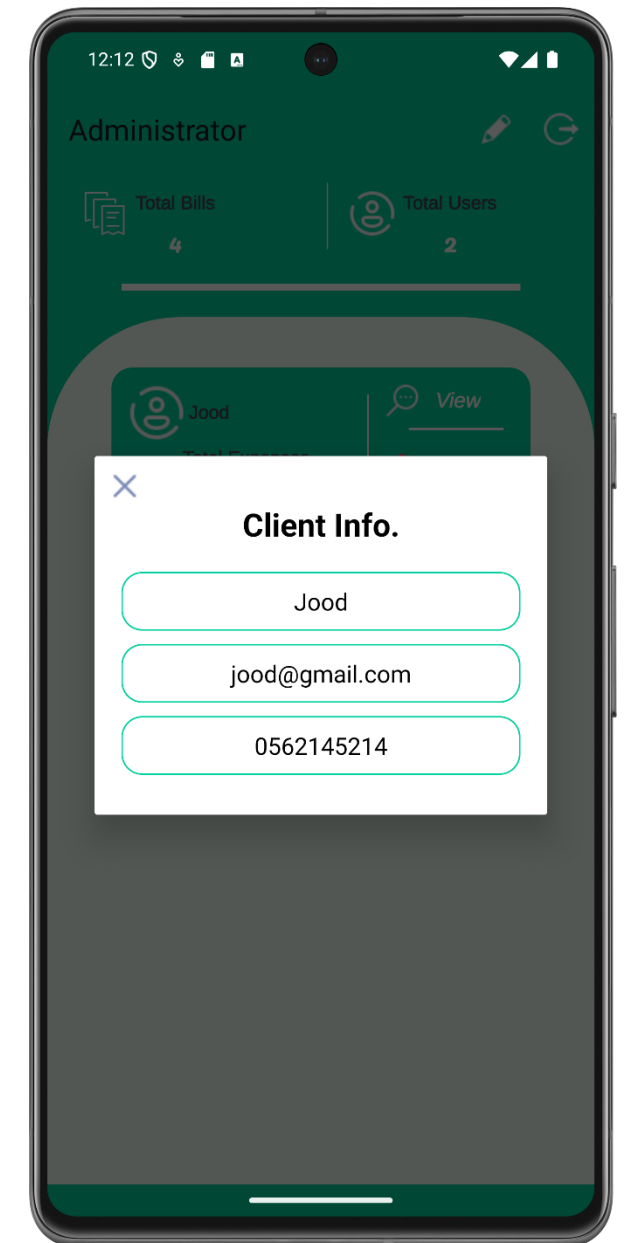
Logout interface

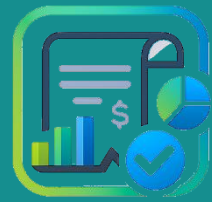


Delete User



View User Details

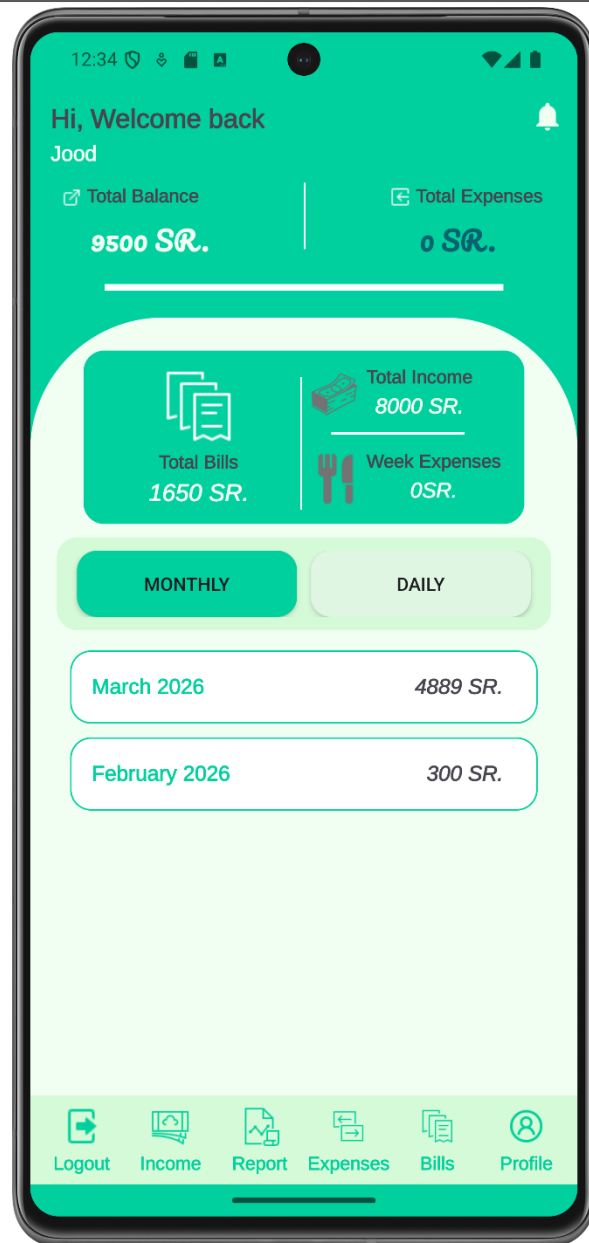




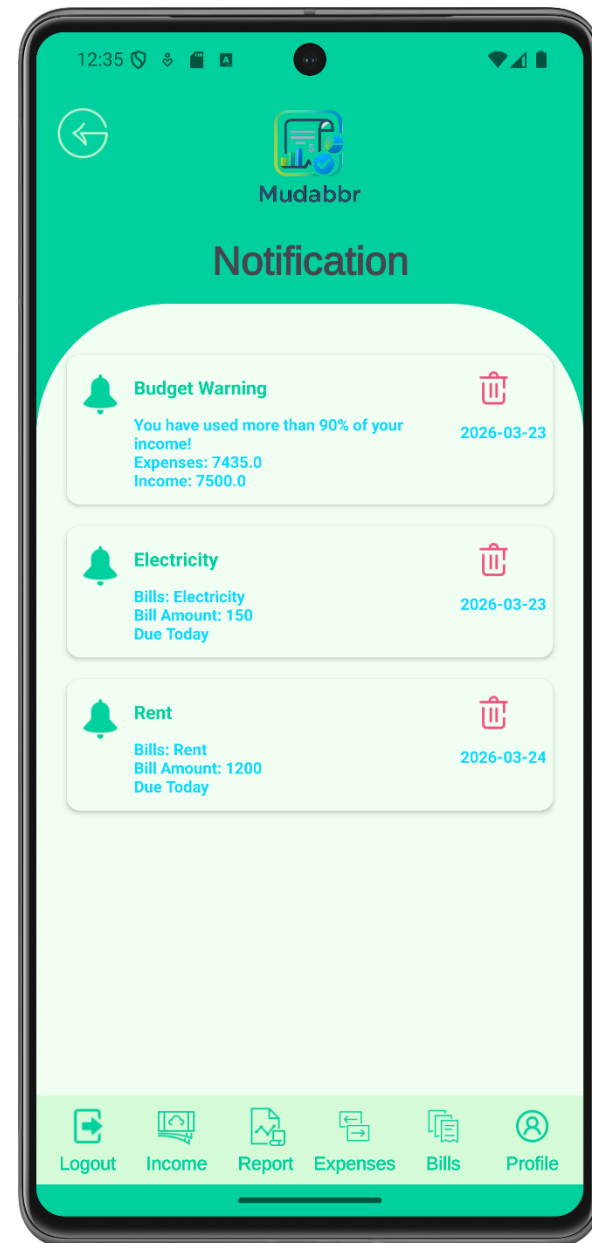
Mudabbr

System Implementation

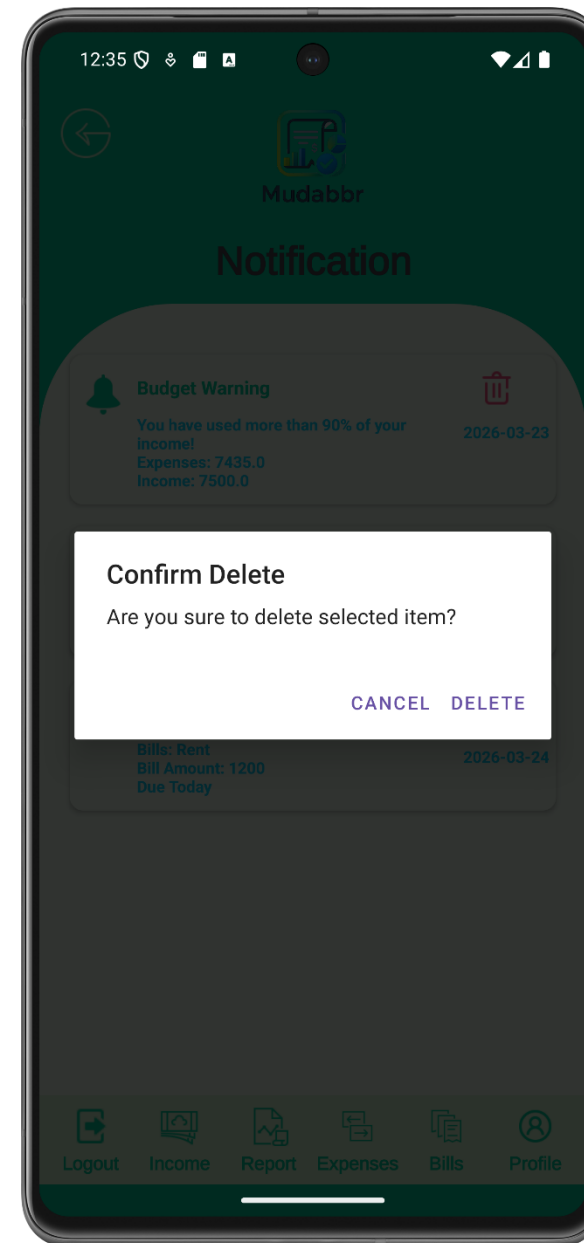
User interface



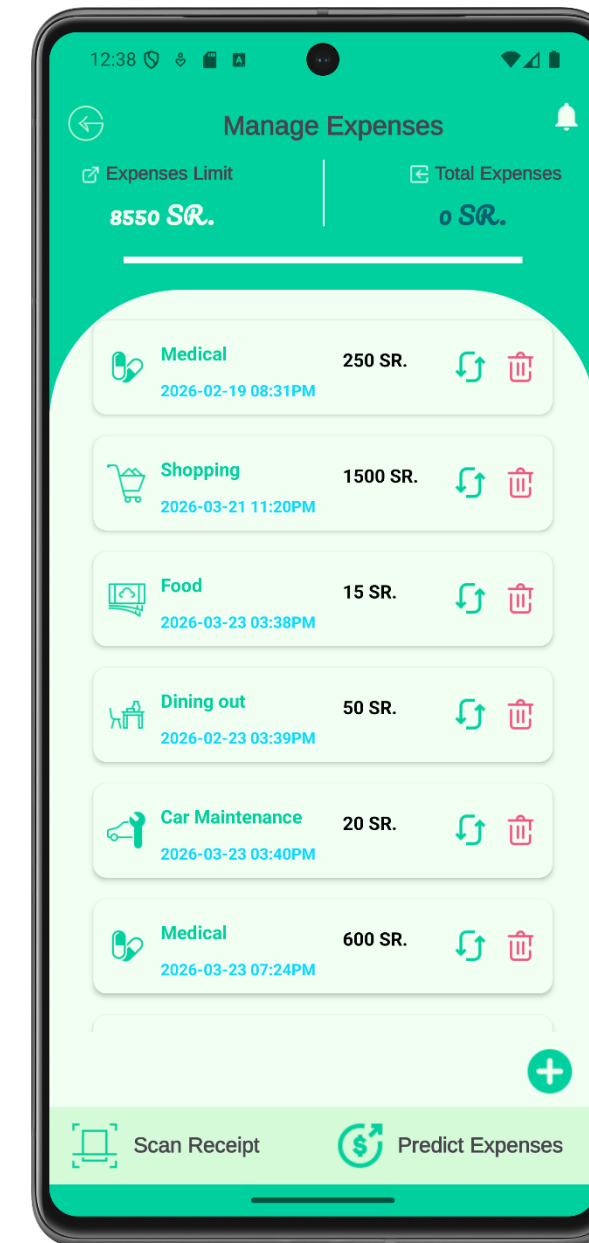
View Notification



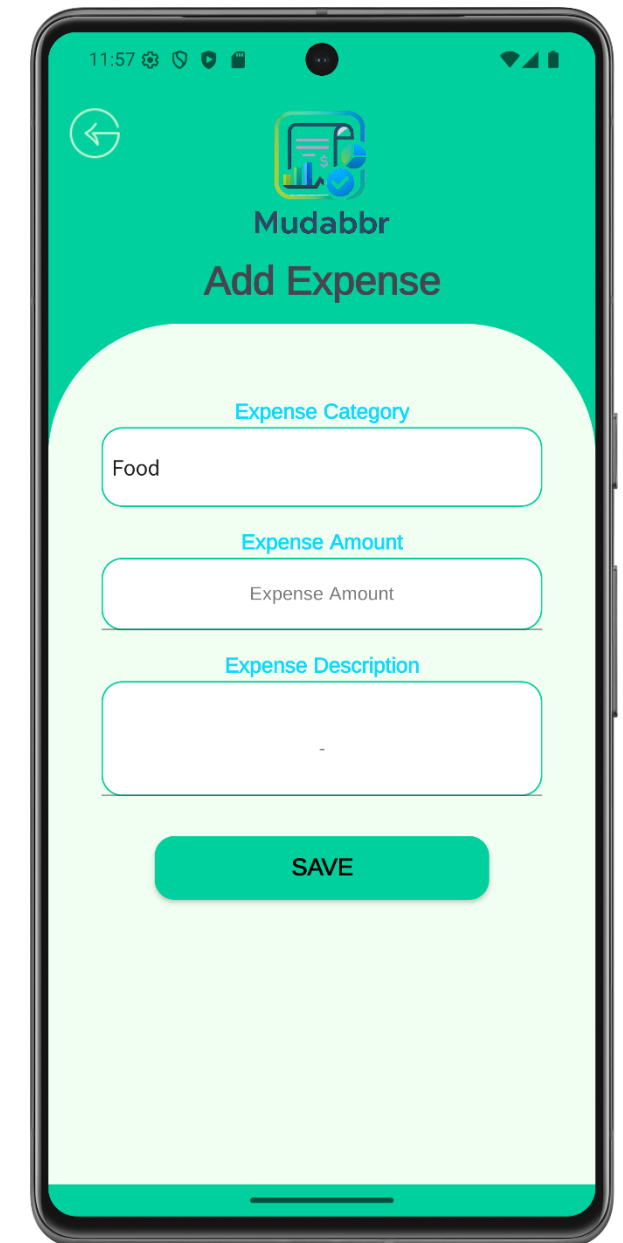
Delete Notification

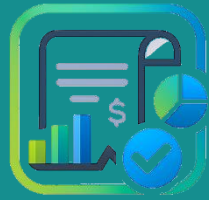


Manage Expenses



Add Expense





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System Implementation

Update Expense

11:57

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Update Expense

Expense Category

Food

Expense Amount

Expense Amount
15 SR.

Expense Description

Tomato

SAVE

Delete Expense

12:38

Manage Expenses

Expenses Limit: 8550 SR. | Total Expenses: 0 SR.

- Medical 250 SR. 2026-02-19 08:31PM
- Shopping 1500 SR. 2026-03-21 11:20PM
- Food 15 SR. 2026-03-23 03:38PM
- Dining out 50 SR. 2026-02-23 03:39PM
- Car Maintenance 20 SR. 2026-03-23 03:40PM
- Medical 600 SR. 2026-03-23 07:24PM

Scan Receipt Predict Expenses

Delete Notification

12:41

Manage Expenses

Expenses Limit: 8550 SR. | Total Expenses: 150 SR.

Confirm Delete

Are you sure to delete selected item?

CANCEL DELETE

Scan Receipt

12:41

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Scan Receipt

START SCAN

Receipt Value

Receipt Description

SAVE

Expense Details

11:56

Manage Expenses

Expenses Limit: 8550 SR. | Total Expenses: 150 SR.

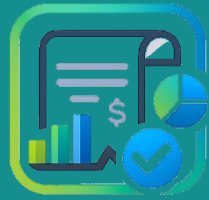
Expense Info.

Medical

2026-02-19 08:31PM

250SR.

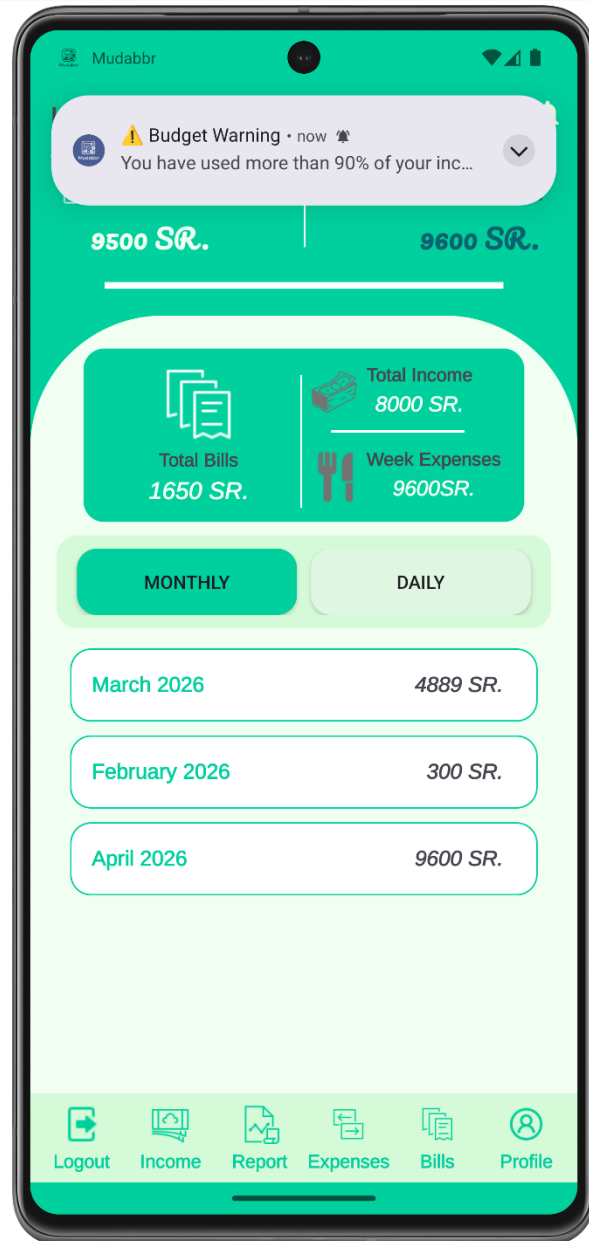
Medicine for my son



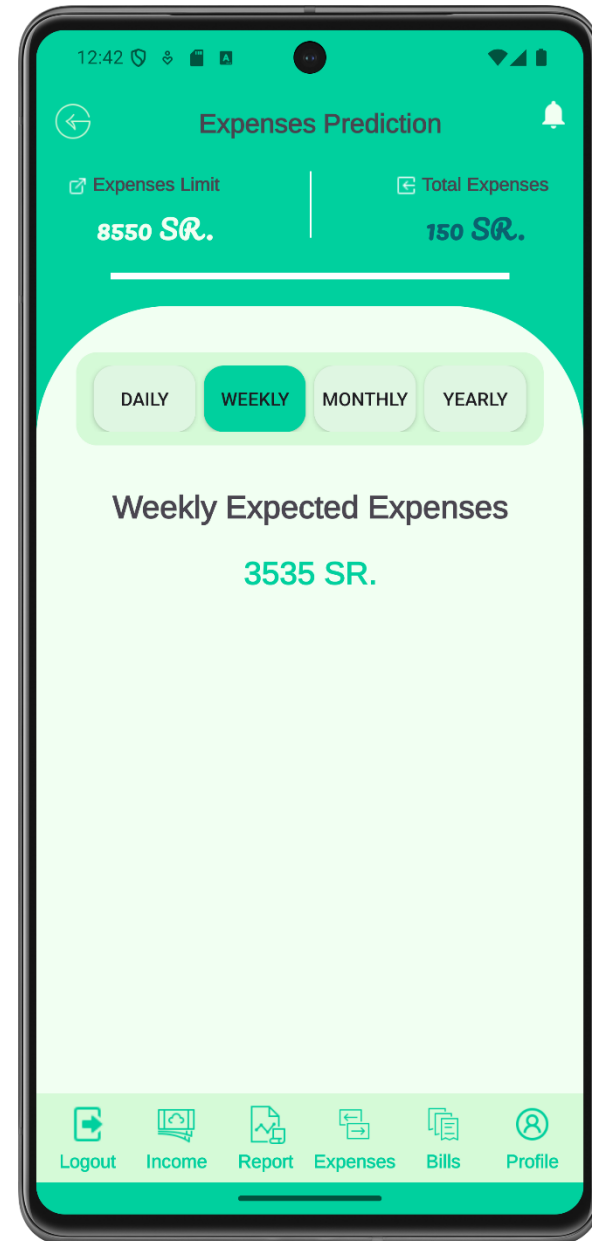
Mudabbr

System Implementation

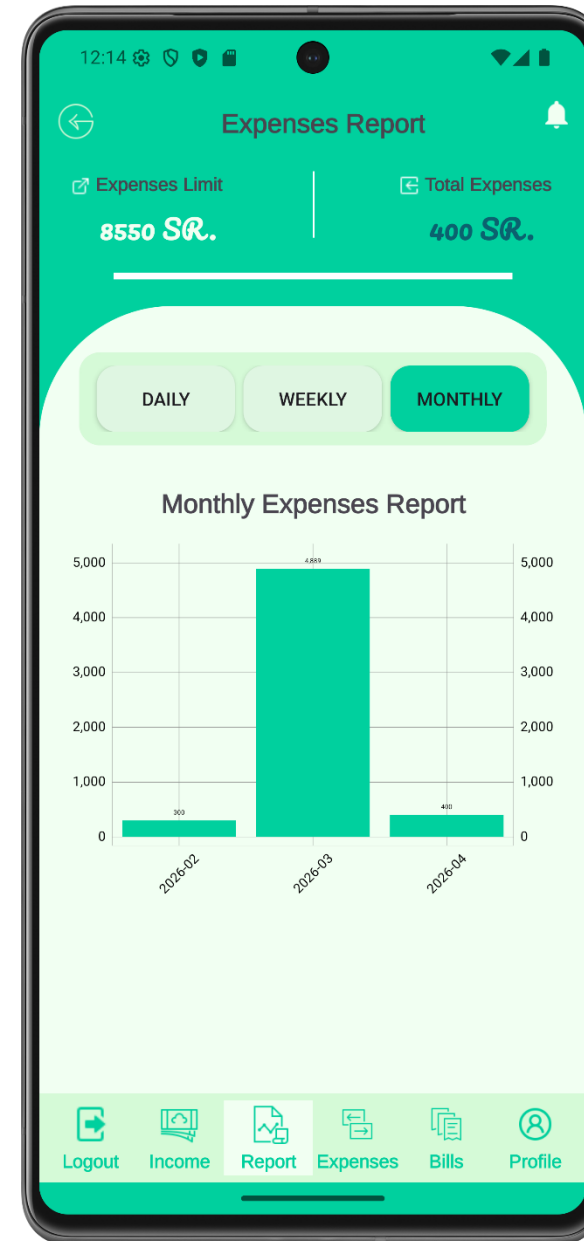
Exceed Budget Alert



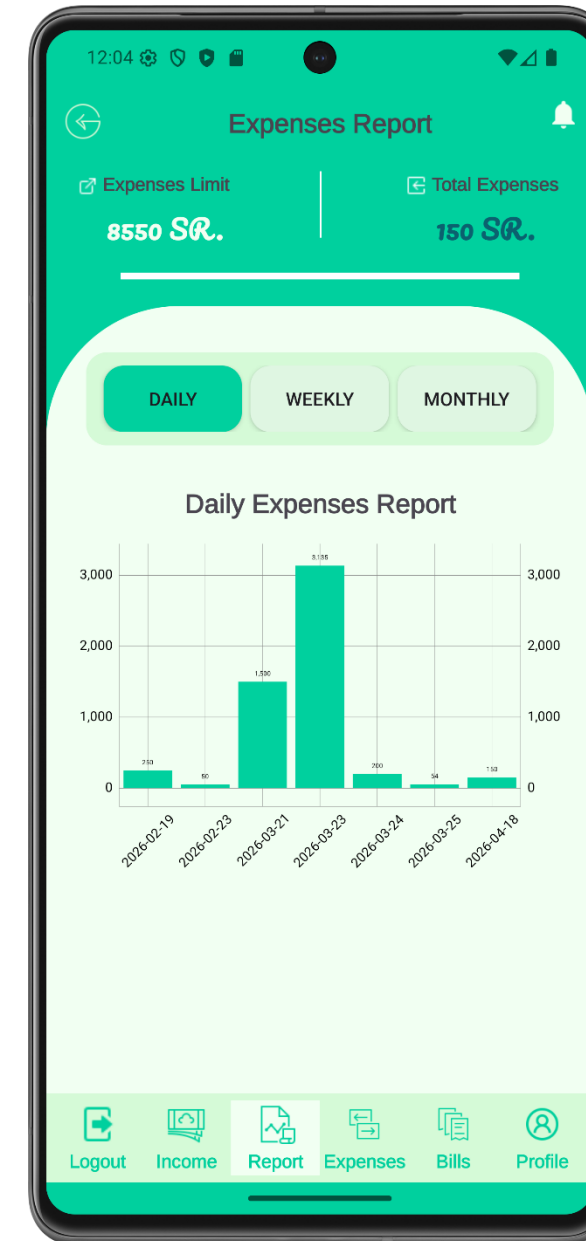
Predict Expense



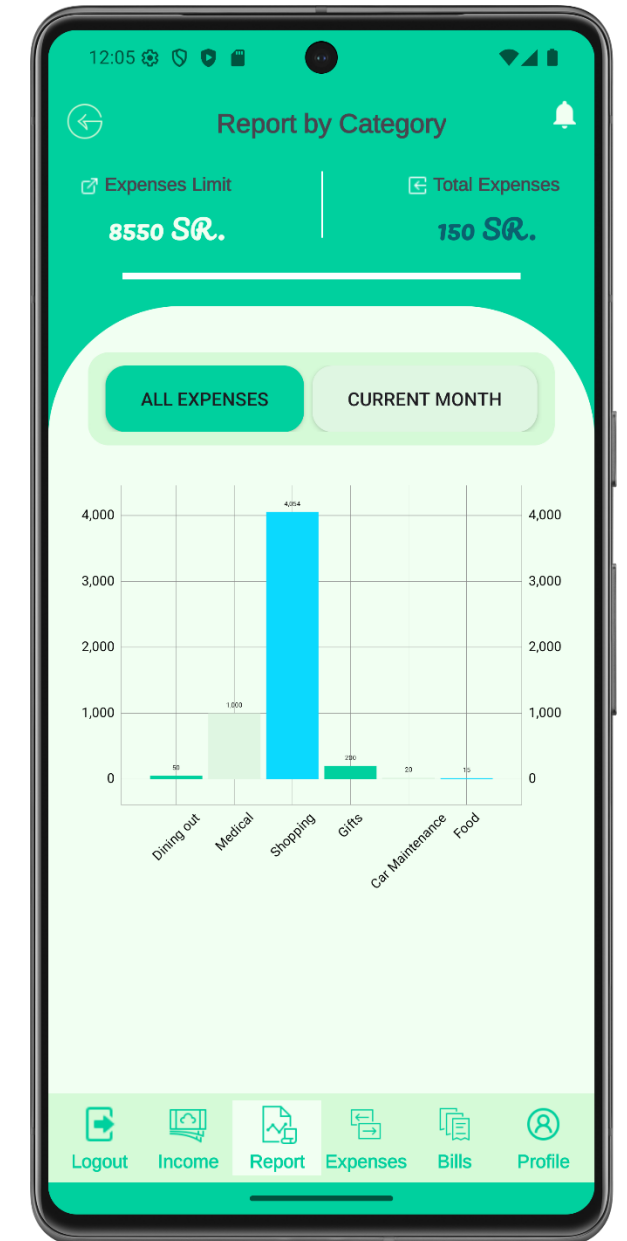
Monthly Expenses

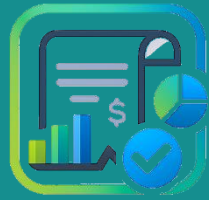


Daily Expenses



Expenses by Category

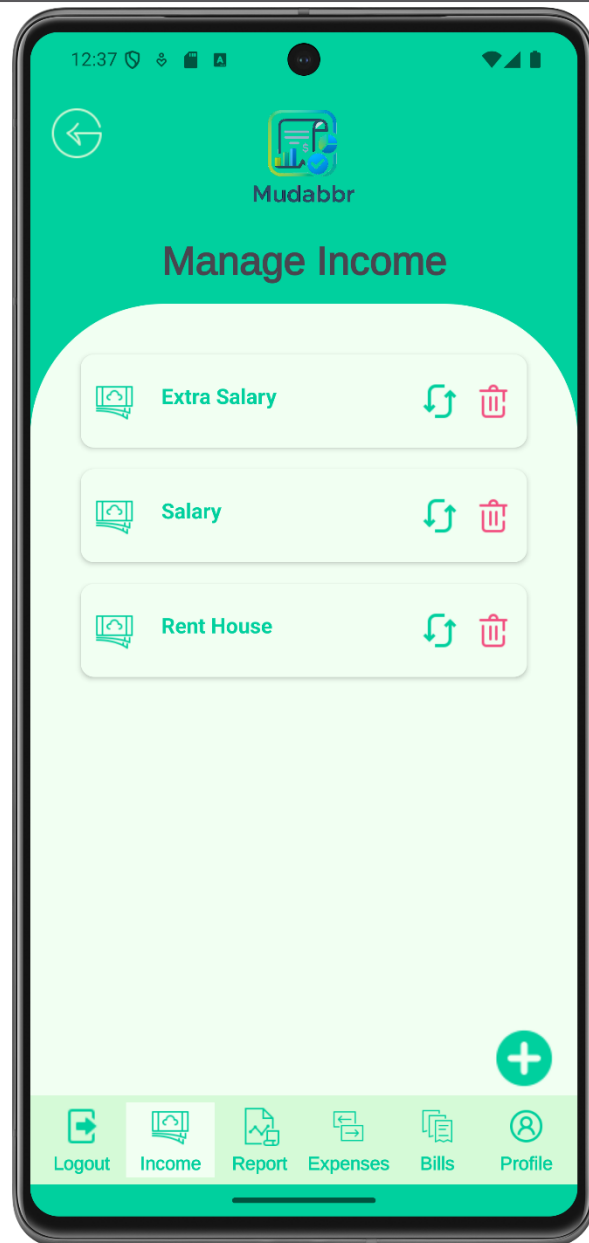




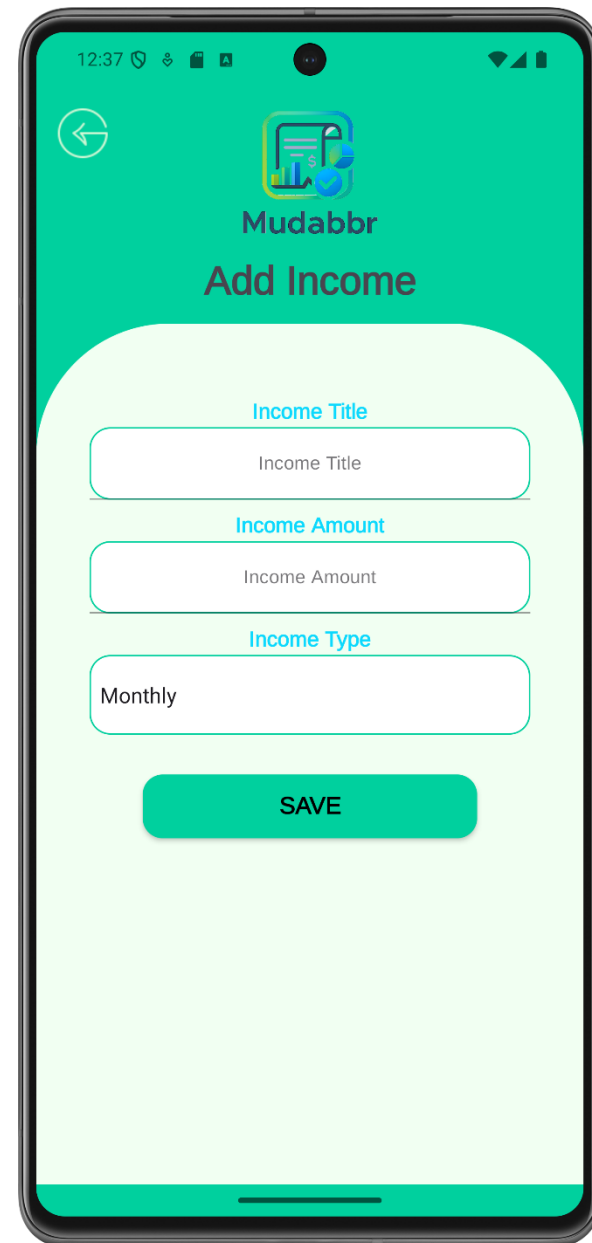
Mudabbr

System Implementation

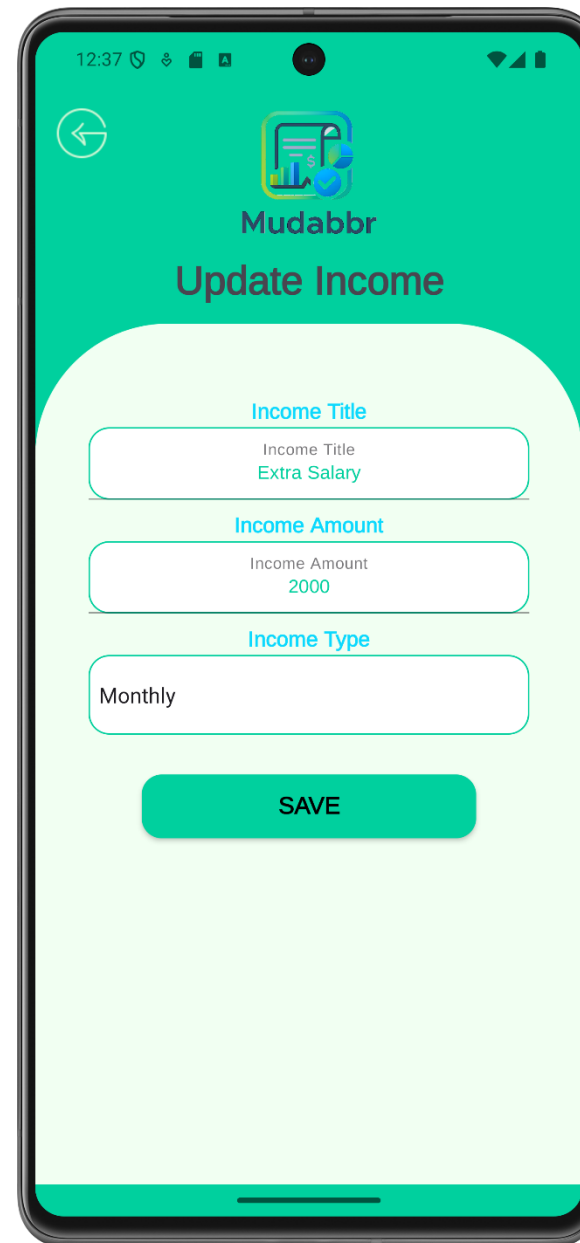
Manage Income



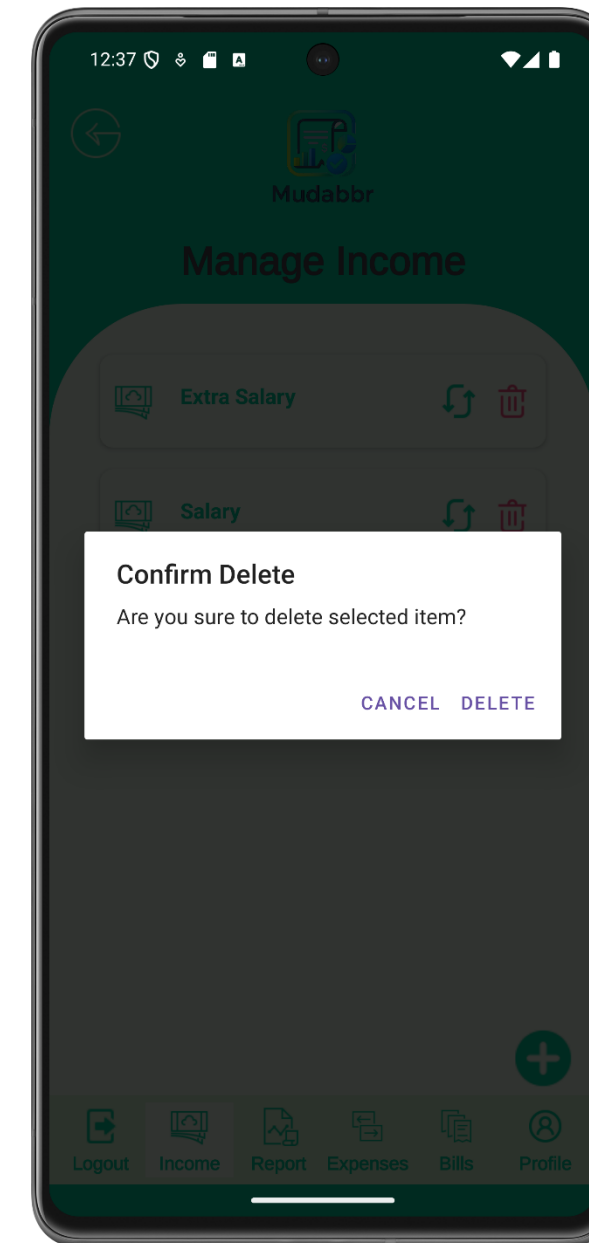
Add Income



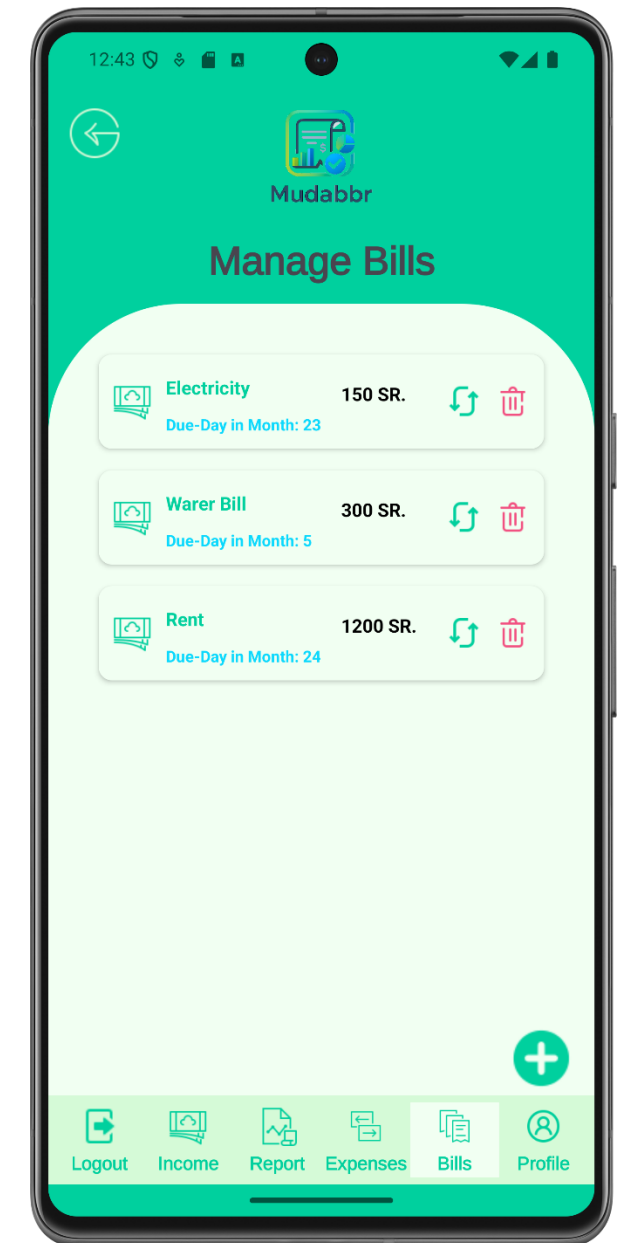
Update Income

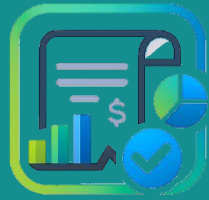


Delete Income



Manage Bills





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System Implementation

Add Bill interface

12:43

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Add Bill

Bill Title

Bill Title

Due Date in Month

Due Date in Month

Bill Amount

Bill Amount

Number of Months

All Months

SAVE

Update Bill

12:49

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Update Bill

Bill Title

Bill Title Electricity

Due Date in Month

Due Date in Month 23

Bill Amount

Bill Amount 150

Number of Months

All Months

Bill State

Wait

SAVE

Delete Bill

12:49

Mudabbr

Manage Bills

Electricity 150 SR. Due-Day in Month: 23

Water Bill 300 SR.

Confirm Delete

Are you sure to delete selected item?

CANCEL DELETE

Logout Income Report Expenses Bills Profile

Bill Details

12:50

Mudabbr

Manage Bills

Bill Info.

Rent

24

1200

Wait

All Months

Logout Income Report Expenses Bills Profile

Bill Due Date Alert

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Bill Reminder • now

Rent Store bill is due today

9500 SR. 2600 SR.

Total Bills 1850 SR.

Total Income 8000 SR.

Week Expenses 2600SR.

MONTHLY DAILY

March 2026 4889 SR.

February 2026 300 SR.

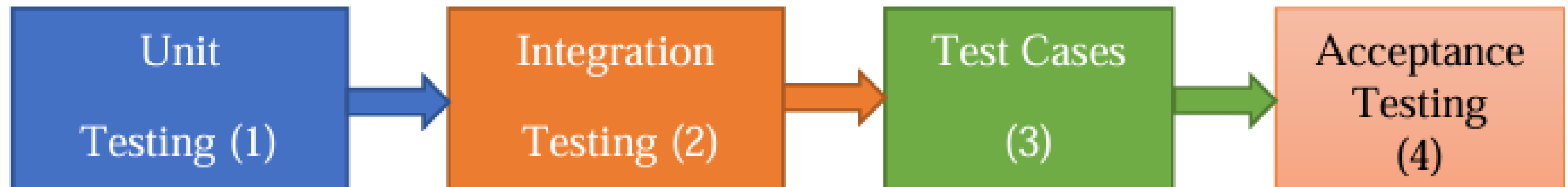
April 2026 2600 SR.

Logout Income Report Expenses Bills Profile



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System Validation – Test Plan





System Validation – Unit Testing

Function to be tested	Test Activity	Pass/Fail
Login	Allow the user to login account	Pass
Register	Allow a new user to create an account	Pass
Forget Password	Allow the user to retrieve the forgotten password	Pass
Reset Password	Allow the user to change account's password	Pass
Logout	Allow the user to exit account	Pass
Edit Profile	Allow the user to view and edit profile	Pass
Login by Finger Print	Allow the user to login using finger print	Pass
Administrator Functions		
View Users	Allow admin to view list of users	Pass
Manage Users	Allow admin to delete any user	Pass
View User Details	Allow the admin to view any user's profile and expenses	Pass
User Functions		
Add Income	Allow the user to add new income	Pass
View Income List	Allow the user to view all his/her income	Pass
Update Income	Allow the user to update his/her income	Pass
Delete Income	Allow the user to delete any of his/her income	Pass
Add Bill	Allow the user to add new bill	Pass



System Validation – Unit Testing

<i>Function to be tested</i>	<i>Test Activity</i>	<i>Pass/Fail</i>
<i>View Bills List</i>	<i>Allow the user to view all his/her bill</i>	<i>Pass</i>
<i>Update Bill</i>	<i>Allow the user to update his/her bill</i>	<i>Pass</i>
<i>Delete Bill</i>	<i>Allow the user to delete any of his/her bill</i>	<i>Pass</i>
<i>Add Expense</i>	<i>Allow the user to add new expense</i>	<i>Pass</i>
<i>View Expense List</i>	<i>Allow the user to view all his/her expense</i>	<i>Pass</i>
<i>Update Expense</i>	<i>Allow the user to update his/her expense</i>	<i>Pass</i>
<i>Delete Expense</i>	<i>Allow the user to delete any of his/her expense</i>	<i>Pass</i>
<i>Scan Receipt & add expenses</i>	<i>Allow the user to scan receipt and adds its value to the user's expenses</i>	<i>Pass</i>
<i>View the Current Expenses</i>	<i>Allow the user to view the current month's total expenses</i>	<i>Pass</i>
<i>Alert User (Expenses exceeds Budget Limit)</i>	<i>The user receives a notification when his/her expenses exceed budget limit</i>	<i>Pass</i>
<i>Alert User (Bill due date)</i>	<i>The user receives a notification when reaching his/her bill due date.</i>	<i>Pass</i>
<i>View Expense Report</i>	<i>The user can preview daily, weekly, and monthly expenses report. And view expenses by category.</i>	<i>Pass</i>
<i>Request Predicted Expenses</i>	<i>Allow the user to view the expected expenses daily, weekly, monthly, and yearly</i>	<i>Pass</i>



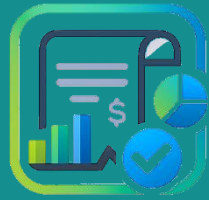
System Validation – Integration Testing

<i>Integrated Functions</i>	<i>Result</i>
<i>Signup → Login → View Profile → Edit Profile → Logout</i>	<i>Done</i>
<i>Admin Login → Users List → View User Details → Delete User → Logout</i>	<i>Done</i>
<i>User Login → Add Expense → View Expenses → Update Expense → Total Expenses updated → Logout</i>	<i>Done</i>
<i>User Login → View Income List → Add New Income → Manage Income</i>	<i>Done</i>
<i>User Login → Scan Receipt → Add to Expenses → Total Expenses Updated</i>	<i>Done</i>
<i>User Login → Add New Bill → Alert User when reach Due Date</i>	<i>Done</i>
<i>User Login → Request Predict Expenses → Predicted Expenses Displayed (Daily, Weekly, Monthly, and Yearly)</i>	<i>Done</i>



System Validation – Acceptance Testing

<i>Requirement</i>	<i>Requirement Description</i>	<i>Status</i>
<i>Security</i>	<i>The application will not be accessible to anyone not authorized to use it through a check email, and password.</i>	<i>Pass</i>
<i>Availability</i>	<i>This system can be accessed 24 /7, and it is accessible by users at any location, and serves many users simultaneously.</i>	<i>Pass</i>
<i>User Friendly Interfaces</i>	<i>Mudabbr App is user-friendly, uses appropriate colors, it requires little training, it is very simple with clear texts.</i>	<i>Pass</i>
<i>Performance</i>	<i>The application's response time is fast.</i>	<i>Pass</i>
<i>Reliability</i>	<i>The data transferred through the application is accurate.</i>	<i>Pass</i>



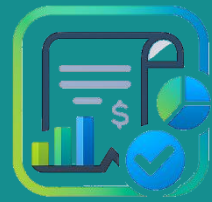
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Future Work



Mudabbr

-  *Developing the system into a website will make it easier for a larger number of users to access it.*
-  *Use advanced machine learning models like LSTM or ensemble methods to more accurately predict user spending.*
-  *Connect with bank accounts, e-wallets, or payment apps to automatically fetch transactions.*
-  *Allow exporting reports as PDF, or Excel for offline analysis.*
-  *Strengthen account security for sensitive financial data.*



Mudabbr

CONCLUSION

Given the increasing financial obligations and challenges individuals face in managing their expenses, the Mudabbr app offers a practical and innovative solution. By OCR technology to automate bill scanning and categorization, the app eliminates the inefficiencies of manual record-keeping and improves the accuracy of expense tracking.

Mudabbr's smart features, such as predictive purchase analysis, monthly budget forecasting, and detailed expense reports, provide users with a deeper understanding of their spending habits and enhance their financial literacy.

Furthermore, the inclusion of biometric and password authentication ensures that security remains a top priority, giving users confidence in managing their sensitive financial data. The Mudabbr app is developed by using Android and Firebase Realtime Database.





Thank You

For your time and attention

Prepared by

Baylsan Hassan Almehdawi

Jood Mohammed Al-Hatemi

Rose Ibraheem Al-Hatemi

Fatima Mohammed Al-Barakati

Etimad Awwad Almalki



Mudabbr Application System



Mudabbr